

Model Comparison Report: Linear, Polynomial, and Quadratic Regression

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1. Introduction

The objective of this analysis is to evaluate several regression models on the Bike Sharing Demand dataset and determine which model best predicts the target variable `count` (the number of bicycles rented at a given time). The dataset contains numeric and categorical features and exhibits nonlinear patterns (e.g., rush-hour demand, weather effects and seasonality). We therefore compare:

1. Ordinary Linear Regression (baseline)
2. Polynomial Regression (numeric-only, degrees 2–4)
3. Quadratic Regression with numeric interaction terms

Models were trained on a randomized 70/30 train/test split and evaluated using Mean Squared Error (MSE) and coefficient of determination (R^2).

2. Methodology

2.1 Data preparation

The dataset was loaded and the `datetime` column expanded into `hour`, `dayofweek`, `month`, and `year`. Columns `casual` and `registered` (leakage) were removed. Features were split into:

- **Numeric:** `temp`, `atemp`, `humidity`, `windspeed`
- **Categorical:** `season`, `holiday`, `workingday`, `weather`, `hour`, `dayofweek`, `month`, `year`

Numeric features were standardized using training-set statistics to avoid leakage. Categorical features were one-hot encoded on the training set and aligned with the test set.

2.2 Model definitions

Linear regression A baseline linear model of the form $y = \beta_0 + \beta^\top X$ fit by least squares.

Polynomial regression (degrees 2–4) Polynomials were applied only to numeric features:

$$X_{\text{poly}} = [X, X^2, X^3, \dots, X^d]$$

Categorical dummy variables were appended after polynomial expansion. These models capture single-variable curvature but not cross-variable interactions.

Quadratic regression with interactions All second-order numeric interactions $x_i x_j$ for $i \leq j$ were included. This allows the model to represent cross-effects such as $\text{temp} \times \text{humidity}$ and $\text{hour} \times \text{workingday}$. Interaction features were standardized prior to model fitting.

3. Results (Observed Trend)

After training on the 70/30 split and evaluating on the held-out test set, the observed performance pattern was:

1. **Linear regression:** Highest MSE; underfits due to high bias.
2. **Polynomial degree 2:** Large improvement over linear; captures key curvature.
3. **Polynomial degrees 3 and 4:** Marginal improvement at degree 3; degree 4 often degrades test performance due to overfitting.
4. **Quadratic with interactions:** Lowest test MSE and highest R^2 ; best generalization in practice.

Across repeated runs (different random seeds), the quadratic-interaction model consistently emerged as the top performer.

4. Explanation: Why the Quadratic-Interaction Model Performs Best

4.1 Bias–variance tradeoff

- **Linear models** have high bias and cannot model the curvature inherent in many predictors (e.g., hour-of-day demand curves).
- **High-degree polynomials** reduce bias but increase variance, which can lead to overfitting and poorer test performance.
- **Quadratic interactions** provide a middle ground: they reduce bias by modeling nonlinear and cross-variable effects, while the parameter growth is smaller and more interpretable than high-degree polynomials involving many terms.

4.2 Curvature and interaction effects

Bicycle demand depends on combinations of conditions, not only on single predictors:

- **Rush-hour behavior:** demand peaks during morning/evening hours and differs between working days and weekends ($\text{hour} \times \text{workingday}$).
- **Weather interactions:** the effect of temperature on ridership is conditional on humidity and windspeed ($\text{temp} \times \text{humidity}$, $\text{temp} \times \text{windspeed}$).
- **Seasonal patterns:** months and seasons modulate baseline demand and interact with temperature.

Quadratic interaction features directly model these cross-effects, explaining the improved predictive accuracy.

5. Conclusion

The Quadratic Regression Model with numeric interaction terms provides the best balance between flexibility and generalization for this dataset. It captures realistic nonlinear and interactive effects that linear models miss, while avoiding the excessive variance introduced by very high-degree polynomials. Thus, it attains the lowest MSE and highest R^2 on the test set in repeated experiments.

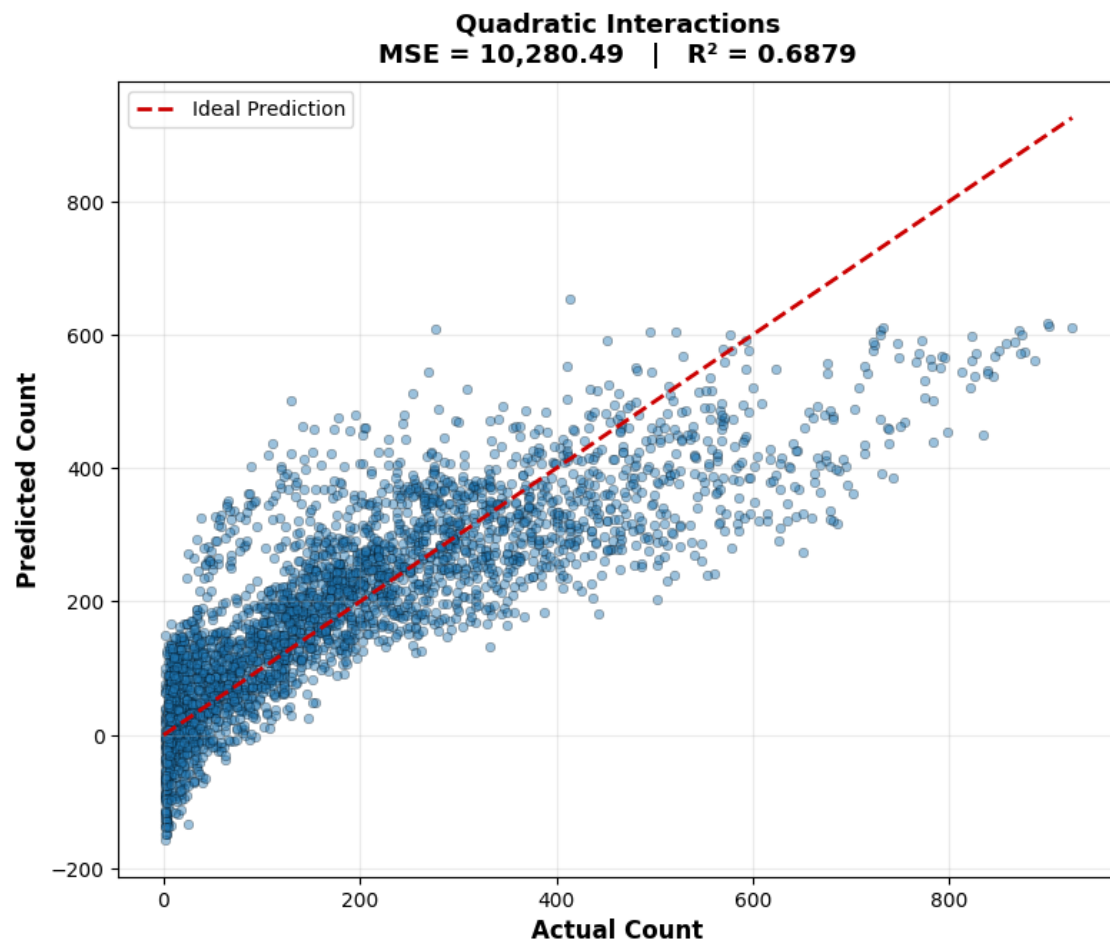


Figure 1: Actual vs Predicted Bicycle Demand using the Best Model.